



KERALA STATE POLLUTION CONTROL BOARD

കേരളസംസ്ഥാന മലിനീകരണ നിയന്ത്രണ ബോർഡ്

Pattom P.O., Thiruvananthapuram – 695 004

പട്ടം പി.ഒ., തിരുവനന്തപുരം - 695 004



File No. PCB/HO/EE3/CERTFN-15/2024

Date: 21/06/2024

Certificate No: PCB/HO/PE/CERTFN/07/2024 dated 21.06.2024.

To

M/s. Sri Venkatalakshmi Spinners (P) Ltd.
Anthiyur, Pulankinar-642122,
Udumalpet(Tk), Tirupur(DT),
Tamil Nadu

Sub: Certificate to Manufacturer - M/s. Sri Venkatalakshmi Spinners (P) Ltd.
Anthiyur, Pulankinar-642122, Udumalpet(Tk), Tirupur(DT), Tamil Nadu
for certification of material for banner and hoarding.

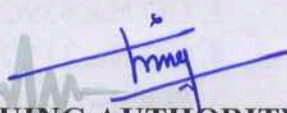
- Ref: 1. Your application no: Nil dated 12/04/2024.
2. Report no. CIPET/KOCHO/CONS.06/2024-25/TC-06 dated 24.05.2024
and CIPET/KOCHO/CONS.07/2024-25/TC-07 dated 24.05.2024
received from CIPET. (Copy enclosed).
3. GO No. 639/2023/LSGD dated 17.03.2023
4. GO No. 111/2019/LSGD dated 29.08.2019
5. SOP No.KSPCB/732/2023-SEE3 dated 10.08.2023 issued by Kerala
State Pollution Control Board
6. Erratum on Standard Operating Procedure on action to be taken in case of
permitted and banned items used for hoarding dated 28.02.2024

With reference to the application received from M/s. Sri Venkatalakshmi Spinners (P) Ltd, whose registered address is at Anthiyur, Pulankinar-642122, Udumalpet(Tk), Tirupur(DT), Tamil Nadu is fulfilling the standard stipulated in GOs and SOP referred above.

Certificate to M/s. Sri Venkatalakshmi Spinners (P) Ltd. Anthiyur, Pulankinar-642122, Udumalpet(Tk), Tirupur(DT), Tamil Nadu for manufacturing

polyethylene material for banner and hoarding material is subject to following conditions.

1. Each material shall have the information related to name and certificate number of manufacturer printed in English.
2. A QR code to the Certificate issued by the Board shall be prepared by M/s. Sri Venkatalakshmi Spinners (P) Ltd, which on scanning has to be lead detailing the contents of this Certificate uploaded in the Board's website. The manufacturer shall provide this QR code to the distributors and printer units also.
3. The details of the QR code (email/password, Name, plant address, KSPCB certificate no. etc.) shall be shared with the printer units and KSPCB (Head Office).
4. If the certified manufacturer is found non-complying any conditions, the Certificate shall stand cancelled.
5. The manufacturer shall provide six-monthly report, giving details of material sale viz. name of printing unit, district, date of issue etc.
6. PCB may cancel or suspend the certificate of the manufacturer, in case the manufacturer fails to comply with any of the conditions of the certificate.

 
ISSUING AUTHORITY
(Environmental Engineer,
Plastic section, Head Office)

Copy to:

- 1) All ROs, DOs
- 2) CA to Chairperson and Member Secretary

1264 512
30.05.24 29/5/24

केंद्रीय पेट्रोसायन अभियांत्रिकी एवं
प्रौद्योगिकी संस्थान (सिपेट)

(पूर्व में सेंट्रल इंस्टिट्यूट ऑफ प्लास्टिक्स इंजीनियरिंग एण्ड टेक्नोलॉजी)
इंस्टिट्यूट ऑफ पेट्रोकेमिकल टेक्नोलॉजी (आई.पी.टी.)

रसायन एवं पेट्रोसायन विभाग

रसायन एवं उर्वरक मंत्रालय, भारत सरकार

एच.आई.एल.कॉलोनी, एडयार रोड, Nr. पातालम्

उद्योगमंडल पी.ओ., कोच्चि, केरल - 683 501

फोन : 0484-2547741

ई-मेल : kochi@cipet.gov.in, cipetkochi@gmail.com

वेबसाइट : www.cipet.gov.in

मुख्यालय : गिण्डी, चेन्नई - 600 032



सिपेट CIPET



आज़ादी का
अमृत महोत्सव



CENTRAL INSTITUTE OF PETROCHEMICALS
ENGINEERING & TECHNOLOGY (CIPET)

(Formerly Central Institute of Plastics Engineering & Technology)

INSTITUTE OF PETROCHEMICALS TECHNOLOGY (IPT)

Department of Chemicals & Petrochemicals

Ministry of Chemicals & Fertilizers, Govt. of India

HIL Colony, Edayar Road, Nr. Pathalam

Udyogamandal P.O., Kochi, Kerala - 683 501

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Web : www.cipet.gov.in

Head Office: Guindy, Chennai-600 032



CIPET: IPT-Kochi /Testing/2024-25/TC-071/583

Date: 24.05.2024

To

M/s.KERALA STATE POLLUTION CONTROL BOARD
PATTOM P.O, THIRUVANANTHAPURAM – 695 004

Dear Sir,

Sub.: Report on Recyclability Assessment – Reg.

Ref.: Letter FILE No. PCB/HO/EE3/CERTFN- 15/2024 Dated 24.04.2024.

We are enclosing herewith consultancy report CIPET/KOCHO/CONS. 07/2024-25/TC-07 dated 24.05.2024 on pertaining to the samples submitted to our laboratory.

Please find enclosed herewith the feedback form. Kindly fill and send it back to us.

Thanking you and assuring you our best services.

Yours sincerely,

S. Anand
Joint Director & Head

Encl. as above.

केन्द्र : अहमदाबाद, अमृतसर, औरंगाबाद, अगरतला, बही, बालासोर, बेंगलूर, भोपाल, भुवनेश्वर, चंद्रपुर, चेन्नई, देहरादून, गुवाहाटी, ग्वालियर, हैदराबाद, हजीपुर, हल्दिया, इम्फाल, जयपुर, कोच्चि, कोरवा, लखनऊ, मदुरै, मुरथल, मैसूरु, रायपुर, राँची, वलसाड, वाराणसी, एवं विजयवाडा

Centres : Ahmedabad, Amritsar, Aurangabad, Agartala, Baddi, Balasore, Bengaluru, Bhopal, Bhubaneswar, Chandrapur, Chennai, Dehradun, Guwahati, Gwalior, Hyderabad, Hajipur, Haldia, Imphal, Jaipur, Kochi, Korba, Lucknow, Madurai, Murthal, Mysuru, Rajpur, Ranchi, Valsad, Varanasi & Vijayawada

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CIPET/KOCHO/CONS.07 /2024-25/TC-07

24.05.2024

To

KERALA STATE POLLUTION CONTROL BOARD

Pattom P.O., Thiruvananthapuram

CERTIFICATE OF ANALYSIS

Sample Details (As stated by party) : Material for fixing banner and hoarding
Quantity of sample submitted : 2 kg + 1 No. A3 size (Unprinted)
Name of the applicant : M/s. Sri Venkatalakshmi Spinners (P) Ltd.,
Anthiyur, Pulankinar - 642122
Udumalpet (Tk), Tirupur (DT), Tamilnadu
Brand Name : -
Sampling : By the party

Test results Obtained:

Sample Details	Test details	Recycling Category as per IS 14534 : 2016
		Material Recovery
		Mechanical Recycling
Material for banner and hoarding (As stated by party)	Assessment of recyclability of banner and certify its category as per IS:14534 (2016)	Suitable in the existing format

Detailed technical analysis on the above results is enclosed as Annexure.


Dr. K.P. Bhuvana
Authorized Signatory


Dr. S. Anbudhayanidhi
Authorized Signatory

केंद्र : अहमदाबाद, अमृतसर, औरंगाबाद, अगरतला, बही, बालासोर, बेंगलूर, भोपाल, भुवनेश्वर, चंद्रपुर, चेन्नई, देहरादून, गुवाहाटी, ग्वालियर, हैदराबाद, हजीपुर, हल्दिया, इम्फाल, जयपुर, कोच्चि, कोरबा, लखनऊ, मदुरै, मुरथल, मैसूरु, रायपुर, राँची, वलसाड, वाराणसी, एवं विजयवाड़ा

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REPORT
ON
RECYCLABILITY ASSESSMENT

Submitted
By



सिपेट CIPET

**Central Institute of Petrochemicals Engineering and
Technology (CIPET): Institute of Petrochemicals
Technology (IPT)**

HIL Colony, Edayar Road,
Pathalam, Eloor, Udyogamandal P.O.,
Kochi-683 501

Web: www.cipet.gov.in, E-mail: kochi@cipet.gov.in

Submitted
to

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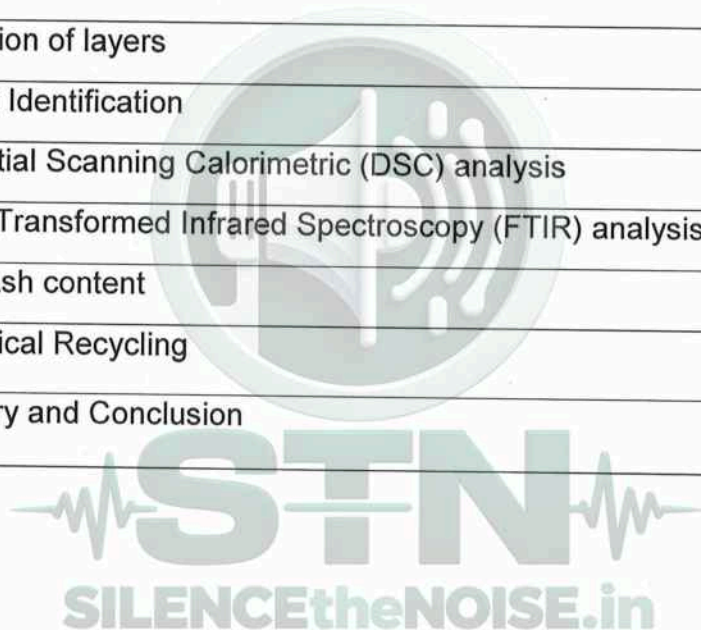


KERALA STATE POLLUTION CONTROL BOARD
Pattom P.O., Thiruvananthapuram

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DETAILED TECHNICAL ANALYSIS REPORT

1. PROJECT OUTLINE:

The scope of the work is to evaluate the suitability of the submitted sample for recycling / energy recovery. The sample was subjected to analytical characterization like Differential Scanning Calorimeter (DSC)/ Fourier Transformed Infrared Spectroscopy (FTIR) to identify the materials and further assessments to study the flow behavior of the sample for recycling and the suitability for energy recovery. The samples were subjected to mechanical recycling and the properties of the recyclates were estimated.

2. DEFINITIONS AS PER IS 14534: 2016

Recycling: Processing of plastics waste materials for the original purpose or for other purposes excluding energy recovery.

Material Recovery: Material processing operations including mechanical recycling, feedstock (Chemical) recycling and organic recycling, but excluding energy recovery.

Mechanical Recycling: Processing of plastics waste into secondary raw material or products without significantly changing the chemical structure of the material.

Organic Recycling: Controlled microbiological treatment of biodegradable plastics waste under aerobic and anaerobic conditions.

Feedstock Recycling: Conversion to monomer or production of new raw materials by changing the chemical structure of plastics waste through cracking, gasification, or depolymerization, excluding energy recovery and incineration.

Biological recycling: Aerobic (composting) or anaerobic (digestion) treatment of biodegradable plastics waste under controlled conditions using microorganisms to produce, in the presence of oxygen, stabilized organic residues, carbon dioxide, and water or in the absence of oxygen, stabilized organic residues, methane, carbon dioxide, and water.



3. EXPERIMENTAL:

The submitted sample was subjected to the following preliminary analysis in order to assess the recyclability of film.

- Visual Examination
- Separation of Layers
- Identification of Materials by DSC& FTIR
- Filler/Ash content
- Recycling through Melt Processing

4. RESULTS ANALYSIS

4.1 Visual Examination

S. No.	Test	Test Method / Test Standard	Observations
1.	Visual Examination	-	• The submitted is white woven fabric having coating on one side and having lamination film on other side • Sample does not have any printed contents • The submitted sample is white in colour
2.	Overall Mass (GSM)	IS 1964	140.8 g/m ²

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4.2 Separation of layers

S. No.	Test	Test Method / Test Standard	Observations
1.	Separation of Layers	Manual separation	The sample comprise of a thin layer of transparent film, a layer of coating and a woven fabric.



4.3 Material Identification

The submitted sample was subjected to DSC and FTIR analysis in order to identify the material of construction in the sample.

4.3.1 Differential Scanning Calorimetry (DSC):

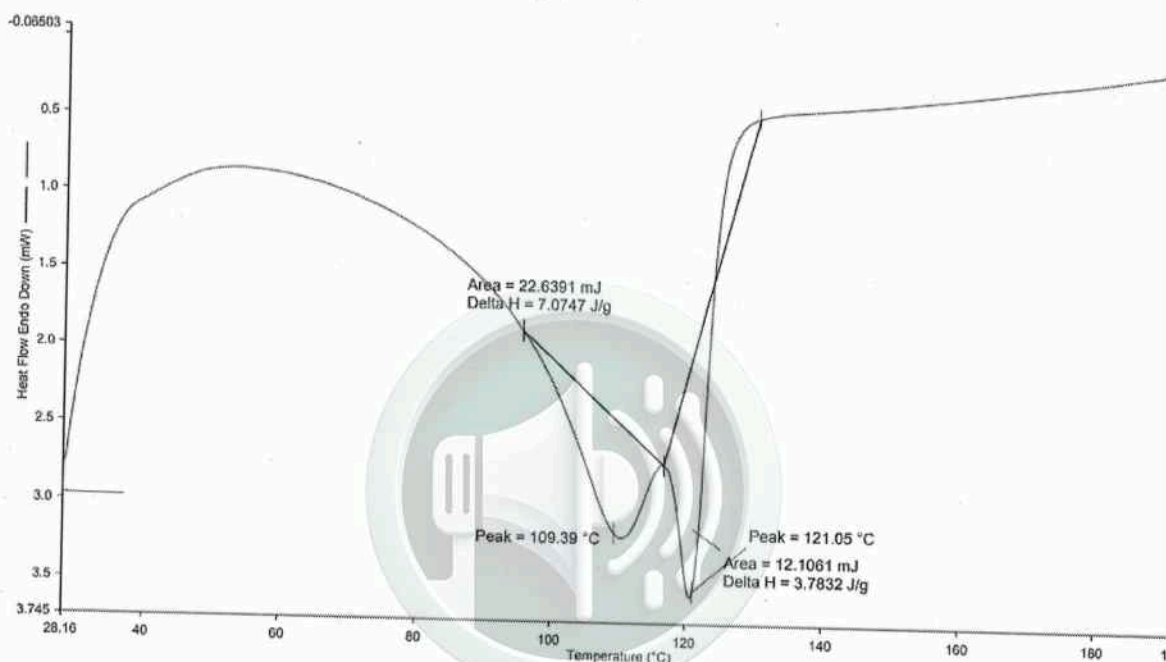


Fig.1. DSC Thermogram of the submitted sample (Laminated film Layer)

The separated laminated film was subjected to DSC analysis in the temperature range of ambient to 200°C at the heating rate of 10°C / min. Fig. 1 depicts the DSC thermogram of the sample. The endothermic peak at 109.39°C and 121.05°C reveals the melting temperature of Low Density Polyethylene (LDPE) and Linear Low Density Polyethylene respectively (LLDPE).



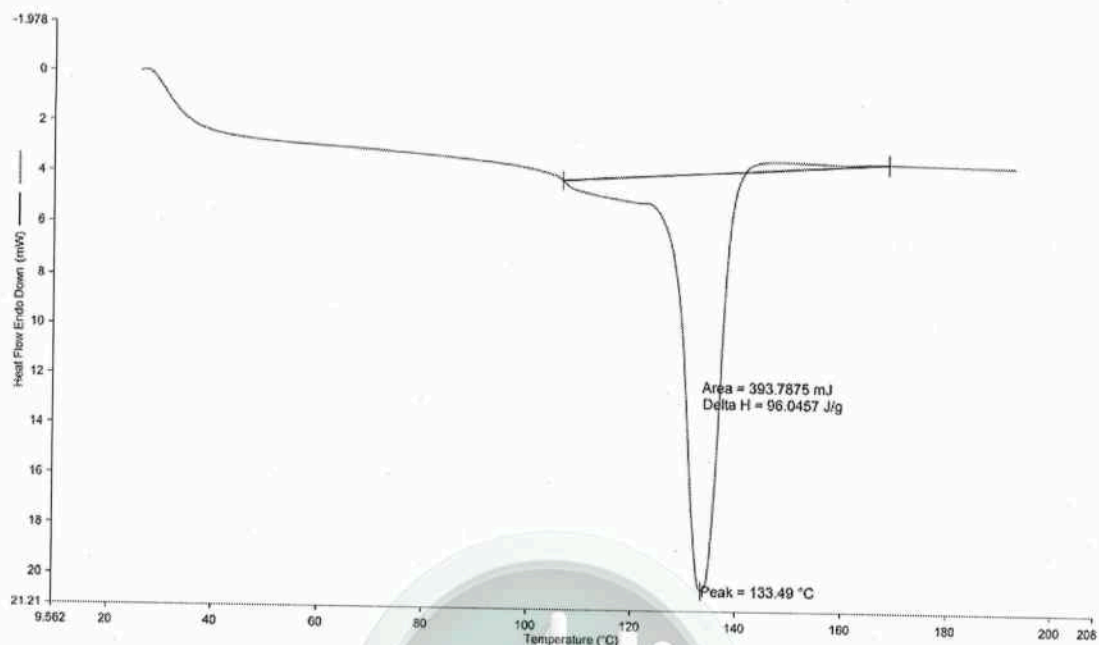


Fig.2. DSC Thermogram of the submitted sample (Tape)

The tape (Yarn) was subjected to DSC analysis in the temperature range of ambient to 200°C at the heating rate of 10°C / min. Fig. 2 depicts the DSC thermogram of the sample. The endothermic peak at 133.49°C the melting temperature of High Density Polyethylene (HDPE).

S. No.	Test	Test Method	Test results	Inference
Tape (Yarn)				
1.	Differential Scanning Calorimetry (DSC)	ASTM D 3418	An endothermic Peaks at 133.49°C	Melting temperature of High Density Polyethylene (HDPE)



Laminated film layer				
1.	Differential Scanning Calorimetry (DSC)	ASTM D 3418	An endothermic Peaks at 109.39°C and 121.05°C	Melting temperature of Low Density Polyethylene (LDPE) and Linear Low Density Polyethylene (LLDPE)

4.3.2 Fourier Transformed Infrared Spectrum

The submitted sample was subjected to Fourier Transformed Infrared (FTIR) studies on both sides.

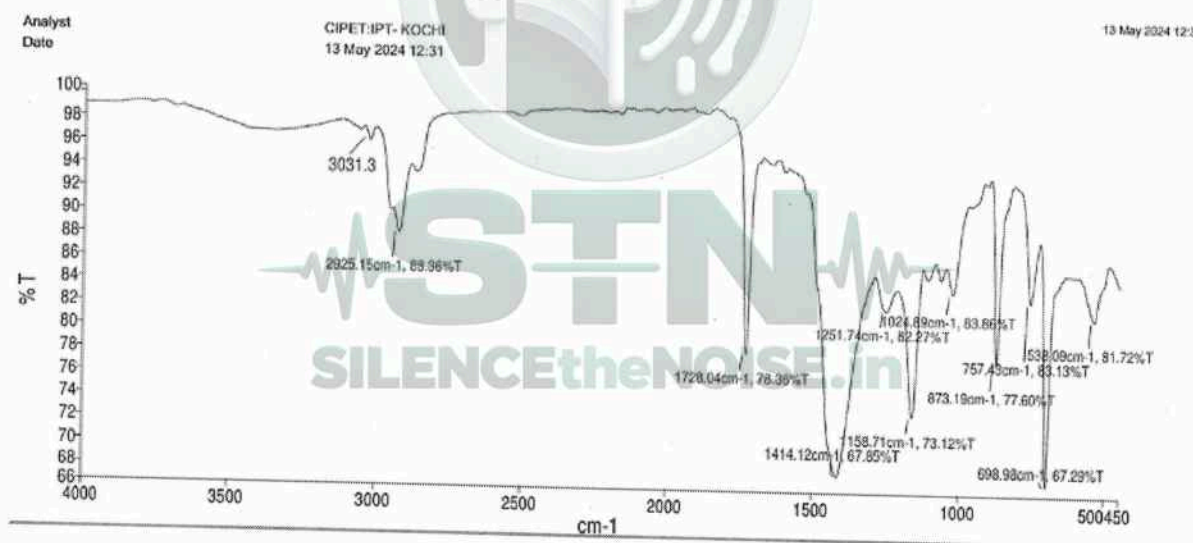


Fig. 3: FTIR spectrum of the submitted sample (Coating)

Inference:

Fig. 2 shows the FTIR spectrum observed at the top surface of the sample. From the above FTIR spectrum, the peaks observed at 3031.3 cm⁻¹ (C-H stretching of benzene ring), 1414.12 cm⁻¹ (C=C stretching), and 757.49 cm⁻¹ (C-H stretching), are the

[Handwritten signature and official stamp of GIPET-IPT, KOCHI, GOVT OF INDIA]

characteristic vibrations of Phenyl ring of styrene. The peak at 1158.71 cm^{-1} corresponds to the C-O-C absorption peak of acrylic. The stretching vibration peaks of acrylates arising from C=O is observed at 1728.04 cm^{-1} . Hence, it is inferred that the submitted sample consists of styrene acrylate based coating. Further, no peak corresponds to chlorine is observed. Hence, it is inferred that the sample does not contain PVC.

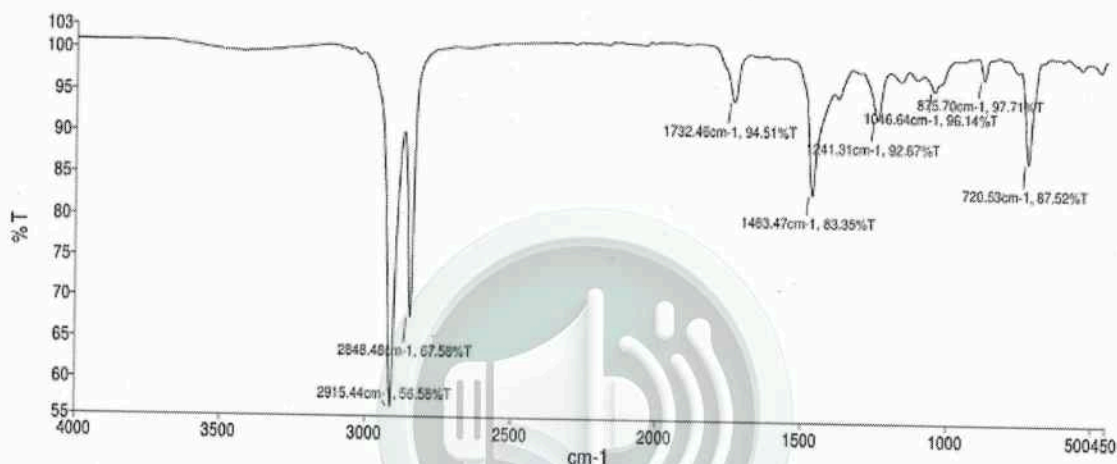


Fig. 4: FTIR spectrum of submitted sample (Laminated Film layer)

Inference:

Fig. 4 shows the FTIR spectrum of separated laminated film layer of the sample. From the above FTIR spectrum the peaks observed at 2915.44 cm^{-1} (CH_2 asymmetric C-H stretching), 2848.48 cm^{-1} (CH_2 symmetric C-H stretching), 1463.47 cm^{-1} (CH_2 deformation), 720.53 cm^{-1} (CH_2 rocking) corresponds to the functional groups of polyethylene. Hence, it is inferred that the submitted sample consists of Polyethylene. Further, no peak corresponds to chlorine is observed. Hence, it is inferred that the sample does not contain PVC.

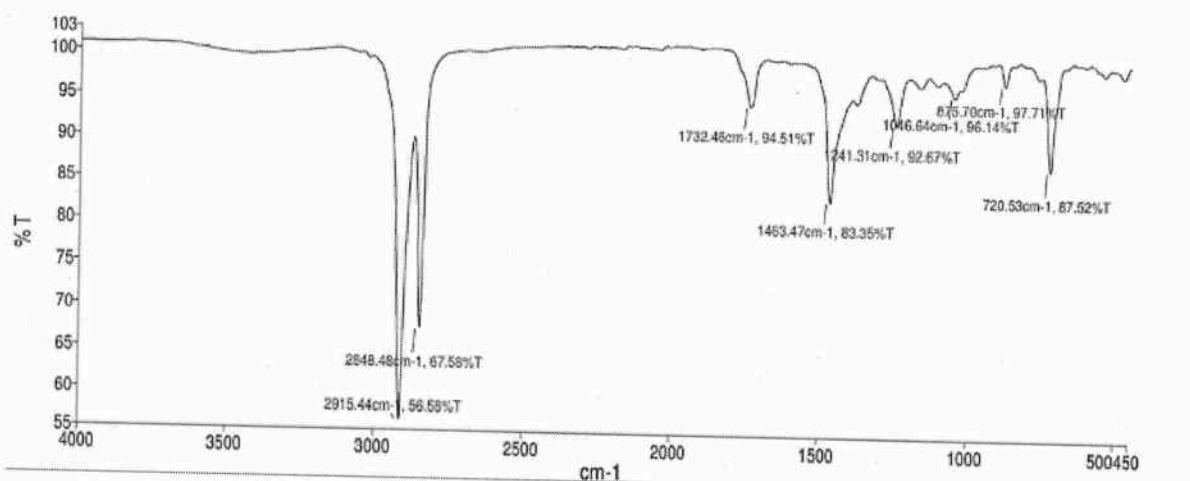


Fig. 5: FTIR spectrum of submitted sample (Tape / Yarn)

Inference:

Fig. 5 shows the FTIR spectrum observed at the Tape / Yarn of the sample. From the above FTIR spectrum the peaks observed at 2913.84 cm^{-1} (CH_2 asymmetric C-H stretching), 2848.48 cm^{-1} (CH_2 symmetric C-H stretching), 1463.47 cm^{-1} (CH_2 deformation), 720.53 cm^{-1} (CH_2 rocking) corresponds to the functional groups of polyethylene. Hence, it is inferred that the submitted sample consist of Polyethylene. Further, no peak corresponds to chlorine is observed. Hence, it is inferred that the sample does not contains PVC.

4.3.3 Filler/Ash Content

The filler/ash content of the submitted sample was tested as per ASTM D 5630. The filler content in the sample was estimated as 15.13%. The filler was identified as calcium carbonate (by chemical method).

4.4. Mechanical Recycling:

4.4.1 Melt processing and granulation

The submitted sample was subjected to mechanical recycling through melt processing technique. The sample in the existing format was fed into the feed zone of the twin screw extruder. The optimal temperature was set in each heating zone of the extruder.

The molten extrudate was subjected to specimen preparation using injection moulding process. The molded specimen was subjected to mechanical testing.





Sample



Recycled Granules

Fig. 6: Recycled Granules from submitted sample

4.4.2 Specimen Preparation

The recycled granules were converted into test specimens using Injection moulding process in order to determine the properties of the recyclates. Fig. 7 shows the molded test specimen from the recycled granules.



Fig. 7: Molded test specimen from recycled granules

The following properties were determined on the recyclate obtained from mechanical recycling of submitted sample.

Sl. No	Name of test	Testing Method	Unit	Test value obtained
01	Tensile strength	ASTM D638	MPa	10.72
02	Elongation at break	ASTM D638	%	74.15
03	Flexural strength	ASTM D790	MPa	13.15
04	Izod Impact strength	ASTM D 256	kJ/m^2 J/m	3.88 39.48
05	Melt flow Index	ASTM D 1238	g/10 min	0.90
06	Filler content	ASTM D 5630	%	15.13
07	Filler Identification	Chemical method	-	Calcium carbonate



5. SUMMARY AND CONCLUSION:

- Visual examination confirms that the submitted sample is in the form of woven fabric, and no printing on either side.
- FTIR and DSC reveal that the submitted sample is constructed with High Density Polyethylene (HDPE) tapes/Yarn (woven fabric), a thin film of blend of Low Density Polyethylene (LDPE) and Linear Low Density Polyethylene (LLDPE) on one side and having coating of styrene acrylate on other side.
- Sample contains 15.13 % of filler. The filler is identified as calcium carbonate.
- Further, there is no evidence of chlorine in the sample, confirming the absence of PVC.
- The melt processing confirms the feasibility of mechanical recycling of the submitted sample of film in the existing format.

Hence, it is concluded that the submitted sample of "Material for banner and hoarding-printed" (**as stated by party**) comprises of High Density Polyethylene (HDPE) tapes/Yarn (woven fabric), a thin film of blend of Low Density Polyethylene (LDPE) and Linear Low Density Polyethylene (LLDPE) on one side, having coating of styrene acrylate on other side and calcium carbonate as filler is suitable for Mechanical recycling (in the existing format) as per IS:14534-2016.

Dr. K.P. Bhuvana
Authorized Signatory

S. An 45
Dr. S. Anbudayanidhi
Authorized Signatory

*****End of the report*****

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रसायन एवं उर्वरक मंत्रालय, भारत सरकार

एच.आई.एल. कॉलोनी, एडयार रोड, Nr. पातालम्

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मुख्यालय : गिण्डी, चेन्नई - 600 032



सिपेट CIPET



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Date: 24.05.2024

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Yours sincerely,

Joint Director & Head

Encl. as above.

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केन्द्र : अहमदाबाद, अमृतसर, औरंगाबाद, अगरतला, बही, बालासोर, बेंगलूर, भोपाल, भुवनेश्वर, चंद्रपुर, चेन्नई, देहरादून, गुवाहाटी, ग्वालियर, हैदराबाद, हजीपुर, हल्दिया, इम्फाल, जयपुर, कोच्चि, कोरबा, लखनऊ, मद्रुरै, मुरधल, मैसूरु, रायपूर, राँची, वलसाड, वाराणसी, एवं विजयवाडा

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केंद्रीय पेट्रोसायन अभियांत्रिकी एवं
प्रौद्योगिकी संस्थान (सिपेट)

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इंस्टिट्यूट ऑफ पेट्रोकेमिकल टेक्नोलॉजी (आई.पी.टी.)

रसायन एवं पेट्रोसायन विभाग

रसायन एवं उर्वरक मंत्रालय, भारत सरकार

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सिपेट CIPET



CENTRAL INSTITUTE OF PETROCHEMICALS
ENGINEERING & TECHNOLOGY (CIPET)

(Formerly Central Institute of Plastics Engineering & Technology)

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CIPET/KOCHO/CONS.06 /2024-25/TC-06

24.05.2024

To

KERALA STATE POLLUTION CONTROL BOARD

Pattom P.O., Thiruvananthapuram

CERTIFICATE OF ANALYSIS

Sample Details (As stated by party)	: Material for fixing banner and hoarding
Quantity of sample submitted	: 2 kg + 1 No. A3 size (Printed)
Name of the applicant	: M/s. Sri Venkatalakshmi Spinners (P) Ltd., Anthiyur, Pulankinar - 642122 Udumalpet (Tk), Tirupur (DT), Tamilnadu
Brand Name	: -
Sampling	: By the party

Test results Obtained:

Sample Details	Test details	Recycling Category as per IS 14534 : 2016
		Material Recovery Mechanical Recycling
Material for banner and hoarding (As stated by party)	Assessment of recyclability of banner and certify its category as per IS:14534 (2016)	Suitable in the existing format

Detailed technical analysis on the above results is enclosed as Annexure.

Dr. K.P. Bhuvana
Authorized Signatory

Dr. S. Anbudhyanidhi
Authorized Signatory

केन्द्र : अहमदाबाद, अमृतसर, औरंगाबाद, अगरतला, बडो, बालासोर, बेंगलूर, भोपाल, भुवनेश्वर, चंद्रपुर, चेन्नई, देहरादून, गुवाहाटी, ग्वालियर, हैदराबाद, हजीपुर, हल्दिया, इम्फाल, जयपुर, कोच्चि, कोरवा, लखनऊ, मदुरै, मुरधल, मैसूरु, रायपुर, राँची, वलसाड, वाराणसी, एवं विजयवाडा

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REPORT
ON
RECYCLABILITY ASSESSMENT

Submitted
By



**Central Institute of Petrochemicals Engineering and
Technology (CIPET): Institute of Petrochemicals
Technology (IPT)**

HIL Colony, Edayar Road,
Pathalam, Eloor, Udyogamandal P.O.,
Kochi-683 501

Web: www.cipet.gov.in, E-mail: kochi@cipet.gov.in

Submitted
to



KERALA STATE POLLUTION CONTROL BOARD
Pattom P.O., Thiruvananthapuram



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DETAILED TECHNICAL ANALYSIS REPORT

1. PROJECT OUTLINE:

The scope of the work is to evaluate the suitability of the submitted sample for recycling / energy recovery. The sample was subjected to analytical characterization like Differential Scanning Calorimeter (DSC)/ Fourier Transformed Infrared Spectroscopy (FTIR) to identify the materials and further assessments to study the flow behavior of the sample for recycling and the suitability for energy recovery. The samples were subjected to mechanical recycling and the properties of the recyclates were estimated.

2. DEFINITIONS AS PER IS 14534: 2016

Recycling: Processing of plastics waste materials for the original purpose or for other purposes excluding energy recovery.

Material Recovery: Material processing operations including mechanical recycling, feedstock (Chemical) recycling and organic recycling, but excluding energy recovery.

Mechanical Recycling: Processing of plastics waste into secondary raw material or products without significantly changing the chemical structure of the material.

Organic Recycling: Controlled microbiological treatment of biodegradable plastics waste under aerobic and anaerobic conditions.

Feedstock Recycling: Conversion to monomer or production of new raw materials by changing the chemical structure of plastics waste through cracking, gasification, or depolymerization, excluding energy recovery and incineration.

Biological recycling: Aerobic (composting) or anaerobic (digestion) treatment of biodegradable plastics waste under controlled conditions using microorganisms to produce, in the presence of oxygen, stabilized organic residues, carbon dioxide, and water or in the absence of oxygen, stabilized organic residues, methane, carbon dioxide, and water.



3. EXPERIMENTAL:

The submitted sample was subjected to the following preliminary analysis in order to assess the recyclability of film.

- Visual Examination
- Separation of Layers
- Identification of Materials by DSC& FTIR
- Filler/Ash content
- Recycling through Melt Processing

4. RESULTS ANALYSIS

4.1 Visual Examination

S. No.	Test	Test Method / Test Standard	Observations
1.	Visual Examination	-	• The submitted is white woven fabric having coating on printed side and having lamination film on other side
2.	Overall Mass (GSM)	IS 1964	138.4 g/m ²

4.2 Separation of layers

S. No.	Test	Test Method / Test Standard	Observations
1.	Separation of Layers	Manual separation	The sample comprise of a thin layer of transparent film, a layer of coating and a woven fabric.



4.3 Material Identification

The submitted sample was subjected to DSC and FTIR analysis in order to identify the material of construction in the sample.

4.3.1 Differential Scanning Calorimetry (DSC):

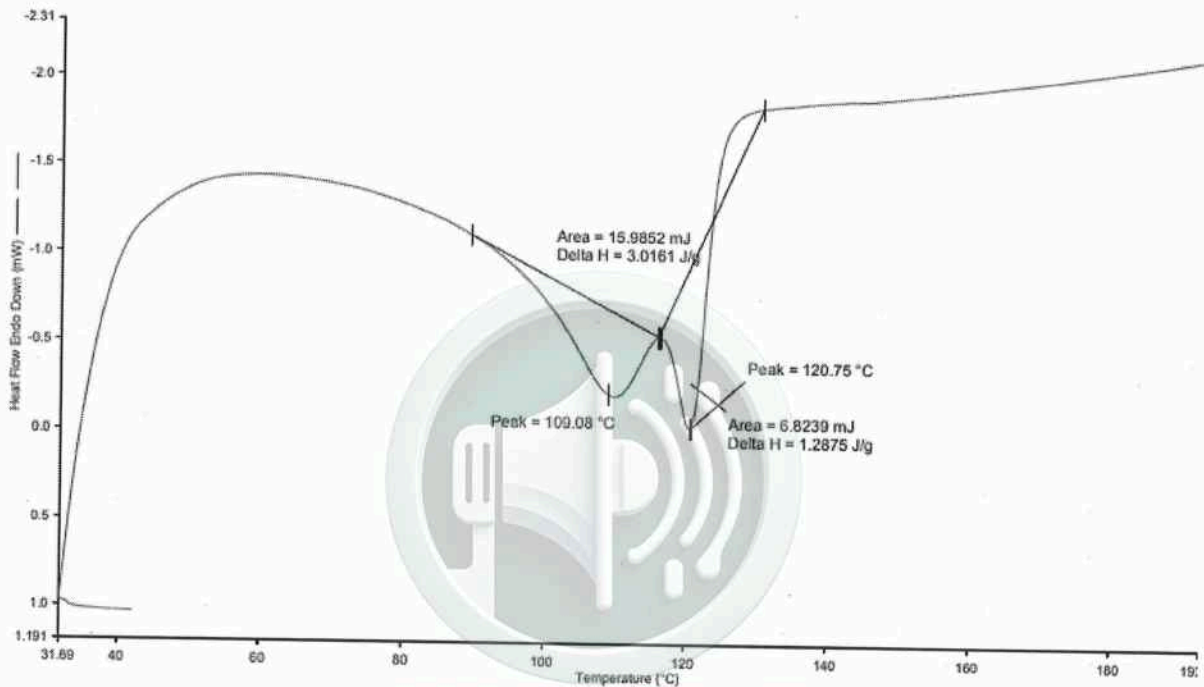


Fig.1. DSC Thermogram of the submitted sample (Laminated film Layer)

The separated laminated film was subjected to DSC analysis in the temperature range of ambient to 200°C at the heating rate of 10°C / min. Fig. 1 depicts the DSC thermogram of the sample. The endothermic peak at 109.08°C and 120.75°C reveals the melting temperature of Low Density Polyethylene (LDPE) and Linear Low Density Polyethylene respectively (LLDPE).



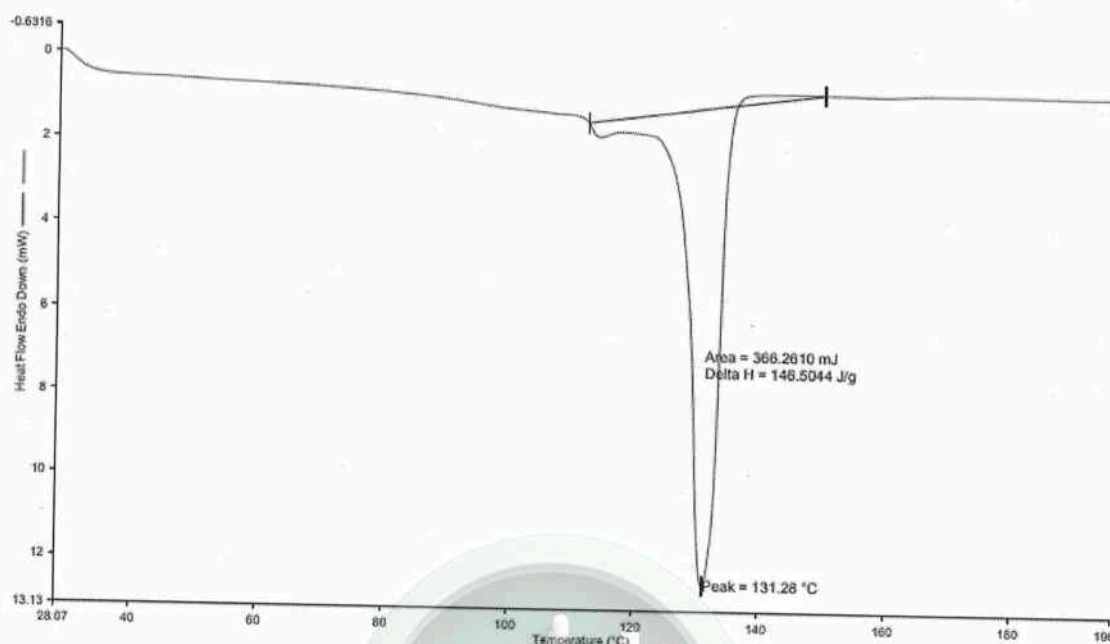


Fig.2. DSC Thermogram of the submitted sample (Tape)

The tape (Yarn) was subjected to DSC analysis in the temperature range of ambient to 200°C at the heating rate of 10°C / min. Fig. 2 depicts the DSC thermogram of the sample. The endothermic peak at 131.28°C the melting temperature of High Density Polyethylene (HDPE).

S. No.	Test	Test Method	Test results	Inference
Tape (Yarn)				
1.	Differential Scanning Calorimetry (DSC)	ASTM D 3418	An endothermic Peaks at 131.28°C	Melting temperature of High Density Polyethylene (HDPE)



Laminated film layer				
1.	Differential Scanning Calorimetry (DSC)	ASTM D 3418	An endothermic Peaks at 109.08°C and 120.75°C	Melting temperature of Low Density Polyethylene (LDPE) and Linear Low Density Polyethylene (LLDPE)

4.3.2 Fourier Transformed Infrared Spectrum

The submitted sample was subjected to Fourier Transformed Infrared (FTIR) studies on both sides.

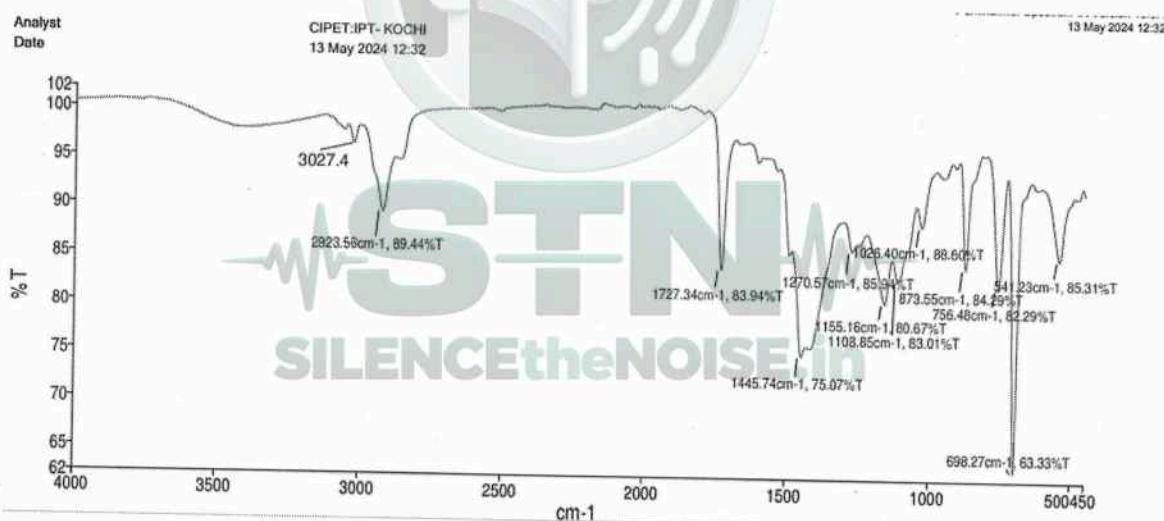


Fig. 3: FTIR spectrum of the submitted sample (Coating)

Inference:

Fig. 2 shows the FTIR spectrum observed at the top surface of the sample. From the above FTIR spectrum, the peaks observed at 3027.4 cm^{-1} (C-H stretching of benzene ring), 1445.7 cm^{-1} (C=C stretching), and 756.48 cm^{-1} (C-H stretching), are the characteristic vibrations of Phenyl ring of styrene. The peak at 1155 cm^{-1} corresponds to the C-O-C absorption peak of acrylic. The stretching vibration peaks of acrylates



arising from C=O is observed at 1727.34 cm^{-1} . Hence, it is inferred that the submitted sample consist of styrene acrylate based coating. Further, no peak corresponds to chlorine is observed. Hence, it is inferred that the sample does not contains PVC.

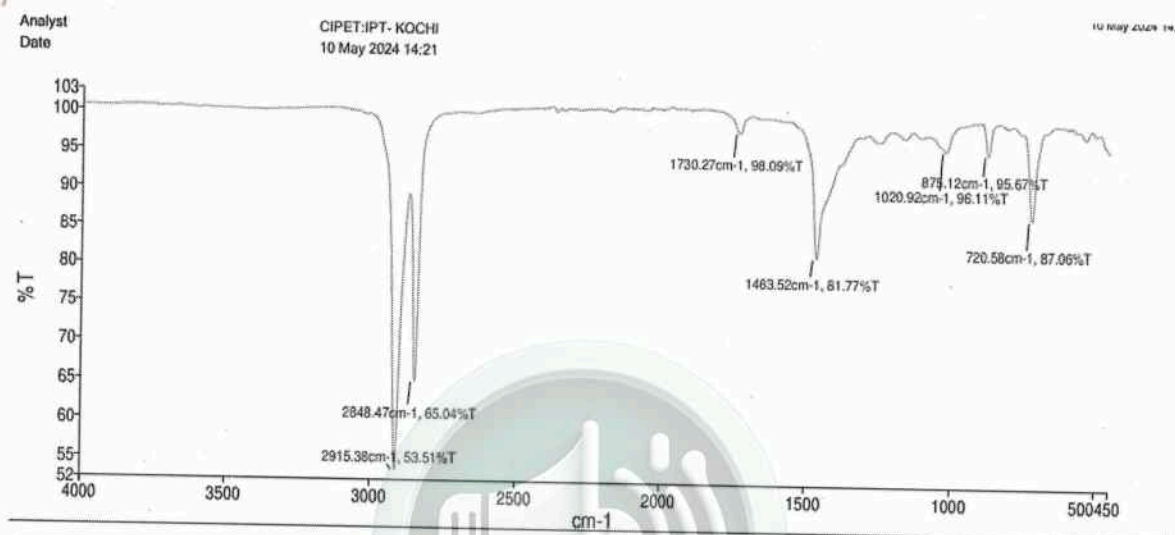


Fig. 4: FTIR spectrum of submitted sample (Laminated Film layer)

Inference:

Fig. 4 shows the FTIR spectrum of separated laminated film layer of the sample. From the above FTIR spectrum the peaks observed at 2915.38 cm^{-1} (CH_2 asymmetric C-H stretching), 2848.47 cm^{-1} (CH_2 symmetric C-H stretching), 1463.52 cm^{-1} (CH_2 deformation), 720.58 cm^{-1} (CH_2 rocking) corresponds to the functional groups of polyethylene. Hence, it is inferred that the submitted sample consist of Polyethylene. Further, no peak corresponds to chlorine is observed. Hence, it is inferred that the sample does not contains PVC.



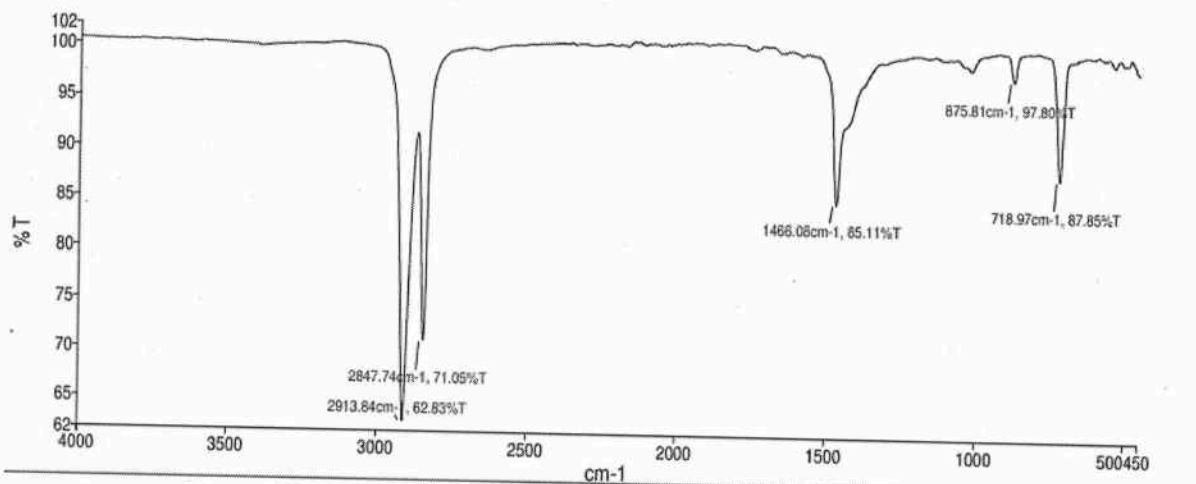


Fig. 5: FTIR spectrum of submitted sample (Tape / Yarn)

Inference:

Fig. 5 shows the FTIR spectrum observed at the Tape / Yarn of the sample. From the above FTIR spectrum the peaks observed at 2913.84 cm^{-1} (CH_2 asymmetric C-H stretching), 2847.74 cm^{-1} (CH_2 symmetric C-H stretching), 1466.08 cm^{-1} (CH_2 deformation), 718.97 cm^{-1} (CH_2 rocking) corresponds to the functional groups of polyethylene. Hence, it is inferred that the submitted sample consist of Polyethylene. Further, no peak corresponds to chlorine is observed. Hence, it is inferred that the sample does not contains PVC.

4.3.3 Filler/Ash Content

The filler/ash content of the submitted sample was tested as per ASTM D 5630. The filler content in the sample was estimated as 16.58%. The filler was identified as calcium carbonate (by chemical method).

4.4. Mechanical Recycling:

4.4.1 Melt processing and granulation

The submitted sample was subjected to mechanical recycling through melt processing technique. The sample in the existing format was fed into the feed zone of the twin screw extruder. The optimal temperature was set in each heating zone of the extruder.

The molten extrudate was subjected to specimen preparation using injection moulding process. The molded specimen was subjected to mechanical testing.



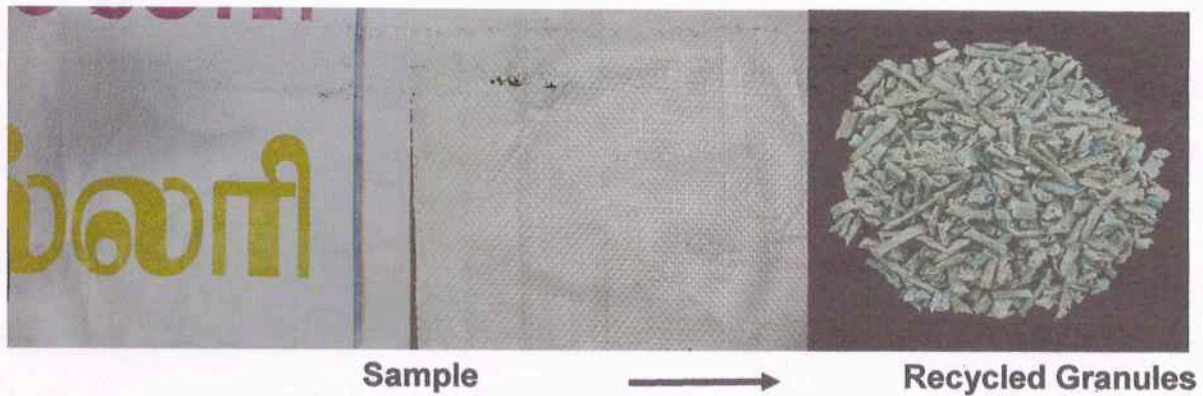


Fig. 6: Recycled Granules from submitted sample

4.4.2 Specimen Preparation

The recycled granules were converted into test specimens using Injection moulding process in order to determine the properties of the recyclates. Fig. 7 shows the molded test specimen from the recycled granules.



Fig. 7: Molded test specimen from recycled granules

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The following properties were determined on the recycle obtained from mechanical recycling of submitted sample.

Sl. No	Name of test	Testing Method	Unit	Test value obtained
01	Tensile strength	ASTM D638	MPa	11.35
02	Elongation at break	ASTM D638	%	210
03	Flexural strength	ASTM D790	MPa	14.52
04	Izod Impact strength	ASTM D 256	kJ/m^2 J/m	4.17 42.36
05	Melt flow Index	ASTM D 1238	g/10 min	0.89
06	Filler content	ASTM D 5630	%	16.58
07	Filler Identification	Chemical method	-	Calcium carbonate



5. SUMMARY AND CONCLUSION:

- Visual examination confirms that the submitted sample is in the form of woven fabric and having printing on one side.
- FTIR and DSC reveal that the submitted sample is constructed with High Density Polyethylene (HDPE) tapes/Yarn (woven fabric), a thin film of blend of Low Density Polyethylene (LDPE) and Linear Low Density Polyethylene (LLDPE) on one side and having coating of styrene acrylate on printed side.
- Sample contains 16.58 % of filler. The filler is identified as calcium carbonate.
- Further, there is no evidence of chlorine in the sample, confirming the absence of PVC.
- The melt processing confirms the feasibility of mechanical recycling of the submitted sample of film in the existing format.

Hence, it is concluded that the submitted sample of "Material for banner and hoarding-printed" (*as stated by party*) comprises of High Density Polyethylene (HDPE) tapes/Yarn (woven fabric), a thin film of blend of Low Density Polyethylene (LDPE) and Linear Low Density Polyethylene (LLDPE) on one side, having coating of styrene acrylate on printed side and calcium carbonate as filler is suitable for Mechanical recycling (in the existing format) as per IS:14534-2016.

Dr. K.P. Bhuvana
Authorized Signatory

S. An 45
Dr. S. Anbudayanidhi
Authorized Signatory

*****End of the report*****

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